Module 3 – Assignment

1. What are the different 802.11 PHY layer standards? Compare their characteristics.

Standard	Wi-Fi Gen	Frequency	Max Data Rate	Modulation	Channel Width	Key Feature
802.11 (legacy)	-	2.4 GHz	2 Mbps	DSSS / FHSS	22 MHz	First standard (1997)
802.11b	Wi-Fi 1	2.4 GHz	11 Mbps	DSSS (CCK)	22 MHz	First widely adopted
802.11a	Wi-Fi 2	5 GHz	54 Mbps	OFDM	20 MHz	Less interference at 5 GHz
802.11g	Wi-Fi 3	2.4 GHz	54 Mbps	OFDM	20 MHz	Backward compatible with 802.11b
802.11n	Wi-Fi 4	2.4 / 5 GHz	600 Mbps	OFDM + MIMO	20/40 MHz	Introduced MIMO
802.11ac	Wi-Fi 5	5 GHz	3.5 Gbps	OFDM + MU-MIMO	20–160 MHz	MU-MIMO, 256-QAM
802.11ax	Wi-Fi 6/6E	2.4/5/6 GHz	9.6 Gbps	OFDMA + MU-MIMO	20–160 MHz	OFDMA, BSS Coloring, TWT
802.11be	Wi-Fi 7	2.4/5/6 GHz	46 Gbps	OFDMA + MU-MIMO	Up to 320 MHz	Multi-Link Operation (MLO)

2. What are DSSS and FHSS? How do they work?

FHSS (Frequency Hopping Spread Spectrum)

FHSS works by "hopping" the carrier signal across a series of sub-channels in a specific, predetermined pattern known only to the transmitter and receiver.

How it works: Imagine a conversation where the two speakers change rooms every 10 seconds. Unless you know the exact sequence of rooms, you cannot follow the conversation.

The Process: The available frequency band is divided into many small channels. The radio transmits a short burst of data on one channel, then jumps to another, then another.

Key Advantage: It is extremely resistant to narrowband interference. If one specific frequency is being jammed (e.g., by a microwave), the signal only loses a tiny fraction of data before it hops to a clean frequency.

Modern Use: While mostly replaced by OFDM in modern Wi-Fi, FHSS is still the core technology behind **Bluetooth**.

2. DSSS (Direct Sequence Spread Spectrum)

DSSS takes a different approach. Instead of jumping between frequencies, it stays on a single wide channel but adds a redundant bit pattern to the data called a **Chipping Code** (or Barker code).

How it works: Every single "bit" of actual data is replaced by a string of bits called "chips." For example, a "1" might be transmitted as "10110111000."

The Process: This makes the signal appear wider and flatter (like noise) across the frequency spectrum.

Key Advantage: Even if part of the signal is damaged by interference during transmission, the receiver can use the mathematical redundancy of the chips to reconstruct the original "1" or "0" without needing a retransmission.

Modern Use: This was the technology that allowed **802.11b** to reach 11 Mbps.

3. How do modulation schemes work in the PHY layer? Compare different modulation schemes and their performance across various Wi-Fi standards.

   As Wi-Fi standards evolved, we moved to higher-order modulation to pack more data into the same amount of radio spectrum.

Modulation	Bits/Symbol	Required SNR	Used In	Notes
BPSK	1	Low	802.11a/b/g/n/ac/ax (low rates)	Most robust, lowest speed
QPSK	2	Low-Medium	802.11a/b/g/n/ac/ax	2x BPSK efficiency
16-QAM	4	Medium	802.11a/g/n/ac/ax	Good balance of speed and reliability
64-QAM	6	Medium-High	802.11n/ac/ax	Used in 802.11n and later
256-QAM	8	High	802.11ac/ax	Introduced in Wi-Fi 5
1024-QAM	10	Very High	802.11ax (Wi-Fi 6)	Requires excellent signal quality
4096-QAM	12	Extremely High	802.11be (Wi-Fi 7)	Maximum throughput, needs near-perfect SNR

   How QAM Works: QAM (Quadrature Amplitude Modulation) varies both the amplitude and phase of the signal. A 256-QAM constellation has 256 distinct points, each representing 8 bits. The AP and client negotiate the highest MCS (Modulation and Coding Scheme) that the signal quality allows. Key Insight: Higher modulation = more bits per symbol = higher speed, but only possible at close range or with excellent signal quality.

4. What is the significance of OFDM in WLAN? How does it improve performance?

   OFDM (Orthogonal Frequency Division Multiplexing) is arguably the most important technology in the history of Wi-Fi. It is the "engine" that allowed Wi-Fi to jump from the slow speeds of 802.11b (11 Mbps) to the high-speed gigabit connections we have today.

   Essentially, OFDM is a method of digital signal modulation where a single high-speed data stream is split into multiple slower-speed "sub-carriers" that are transmitted simultaneously.

1.The Core Concept: Parallel Processing

Think of the difference between a single-lane road and a multi-lane highway.

Single Carrier (Legacy): Data is sent one bit at a time over one wide channel. If a "pothole" (interference) appears on that lane, the whole stream stops.

OFDM (Multi-Carrier): The channel is divided into many small, narrow sub-channels. Data is sent in parallel. Even if one or two sub-channels are blocked by interference, the rest continue to deliver data.

2. Why the "Orthogonal" Part is Key

In traditional radio, if you put two frequencies too close together, they interfere with each other (crosstalk). To prevent this, you usually need "guard bands" (empty space) between frequencies, which wastes spectrum.

OFDM is mathematically designed so that the sub-carriers are **orthogonal** to each other. This means that when one sub-carrier reaches its peak signal, the neighboring sub-carriers are at zero. Because they don't "see" each other, they can be packed extremely tight without any guard bands in between.

3. How OFDM Improves Performance

Resistance to Multipath Distortion:

In an indoor environment, radio waves bounce off walls, furniture, and people. These "echoes" arrive at the receiver at slightly different times, causing **Inter-Symbol Interference (ISI)**.

Improvement: Because OFDM sends data in many small, slow chunks rather than one fast chunk, each symbol lasts longer. This makes the signal much more resilient to these echoes.

5. How are frequency bands divided for Wi-Fi? Explain different bands and their channels.

1. The 2.4 GHz Band

This band is the original home of Wi-Fi. Because of its lower frequency, it has a long range and can pass through solid objects easily. However, it is extremely crowded.

- **Frequency Range:** 2.400 GHz to 2.4835 GHz.
- **Channels:** Divided into 11 channels in North America (13 in most of the world).
- **The Overlap Problem:** Each channel is 20 MHz wide, but the channels are only spaced 5 MHz apart.
- **Non-Overlapping Channels:** To avoid interference, we only use channels **1, 6, and 11**.
- **Best For:** IoT devices (like the TWT sensors we discussed), long-range coverage, and older hardware.

2. The 5 GHz Band

The 5 GHz band was introduced to solve the congestion of 2.4 GHz. It offers much more "spectrum" (space), allowing for wider channels and faster speeds.

- **Frequency Range:** Roughly 5.150 GHz to 5.850 GHz.
- **Channels:** Up to 25 non-overlapping 20 MHz channels.
- **Channel Bonding:** This band allows for "bonding" channels together to create 40 MHz, 80 MHz, or even 160 MHz lanes, which acts like increasing the speed limit on a highway.

- **DFS (Dynamic Frequency Selection):** Some 5 GHz channels are shared with radar (weather/military). If your AP hears radar, it must automatically jump to a different channel.
- **Best For:** Gaming, 4K streaming, and dense office environments.

3. The 6 GHz Band (Wi-Fi 6E and Wi-Fi 7)

This is the newest frontier for Wi-Fi. It opens up a massive amount of clean, "pristine" airwaves because older Wi-Fi devices cannot use it.

- **Frequency Range:** 5.925 GHz to 7.125 GHz.
- **Spectrum Size:** A massive 1,200 MHz of new space.
- **Channel Count:**
  - 59 new 20 MHz channels.
  - Up to 7 new 160 MHz channels.
  - Support for 320 MHz channels in Wi-Fi 7.
- **Zero Interference:** There are no microwaves, baby monitors, or old 802.11b devices here. Every device in this band must support modern security like **WPA3**.
- **Best For:** Ultra-low latency, VR/AR, and multi-gigabit wireless.

6. What is the role of Guard Intervals in WLAN transmission? How does a short Guard Interval improve efficiency?

1. The Problem: Multipath Interference

When an Access Point sends a signal, the radio waves bounce off walls, ceilings, and furniture.

Because of these reflections:

- The "direct" signal arrives first.
- The "reflected" signals (echoes) arrive slightly later.

Without a buffer, the echoes of the **previous symbol** would crash into the beginning of the **current symbol**. This is called **Inter-Symbol Interference (ISI)**. It's like trying to have a conversation in a massive cathedral; if you speak too fast, the echoes of your last word drown out your next one.

2. The Solution: The Guard Interval

The Guard Interval is a period during which the transmitter waits before sending the next symbol. This allows all the "echoes" from the previous symbol to die down so the medium is clear for the new data.

3. Long vs. Short Guard Interval (SGI)

Historically, Wi-Fi used a "Long" Guard Interval to ensure reliability in large, echo-heavy environments. However, modern standards like **802.11n (Wi-Fi 4)** and beyond introduced the **Short Guard Interval (SGI)** to boost performance.

Standard Guard Interval (800 ns)

- Used in older standards and very noisy environments.
- Provides a larger safety net for echoes to settle.
- **Trade-off:** Because we are spending more time "waiting" and less time "talking," the overall data rate is lower.

Short Guard Interval (400 ns)

- Introduced to improve efficiency by cutting the "silence" in half.
- **How it improves efficiency:** By reducing the time wasted between symbols, we can pack more symbols into every second.
- **Result:** Switching from an 800 ns GI to a 400 ns SGI provides an immediate **11% increase in data throughput** without requiring any extra frequency or power.

7. Describe the structure of an 802.11 PHY layer frame. What are its key components?

A standard PHY frame is divided into three distinct functional sections: the **Preamble**, the **Header**, and the **Data (Payload)**.

A. The Preamble (The "Sync" Signal)

Think of the preamble as a loud, rhythmic knocking on a door to get someone's attention. It does not carry "data" in the traditional sense.

- **Synchronization:** It allows the receiver to align its clock with the transmitter's clock.
- **Carrier Sensing:** It tells other devices, "Someone is talking; set your **NAV** (Network Allocation Vector) and stay quiet."
- **Channel Estimation:** It helps the receiver figure out how the environment (walls, interference) is distorting the signal so it can apply a correction filter.

B. The PHY Header (The "Instructions")

Once the receiver is synchronized, it needs to know how to read the rest of the frame. The header contains the "metadata."

- **Rate/MCS:** Tells the receiver which modulation is being used (e.g., **64-QAM** or **BPSK**).
- **Length:** Specifies how long the data portion is, so the receiver knows exactly when the transmission ends.
- **Parity:** A simple check to ensure the header itself wasn't corrupted.

C. The Data / Payload (The "Cargo")

This is the actual **MPDU** (the MAC frame) passed down from the MAC layer. It includes the source/destination MAC addresses and the actual user data (IP packets).

8. What is the difference between OFDM and OFDMA?

Feature	OFDM	OFDMA
Full Form	Orthogonal Frequency Division Multiplexing	Orthogonal Frequency Division Multiple Access
Used In	802.11a/g/n/ac	802.11ax (Wi-Fi 6) and later
Channel Usage	All subcarriers assigned to one user at a time	Subcarriers divided into Resource Units (RUs) shared among multiple users simultaneously
Efficiency	One user occupies entire channel per transmission	Multiple users transmit at the same time on different RUs
Latency	Higher — users must wait for their turn	Lower — concurrent transmissions reduce wait time
Best For	High-throughput single-user scenarios	Dense environments with many simultaneous users (stadiums, offices, airports)
Overhead	Less scheduling complexity	Requires more sophisticated scheduling at AP

9. What is the difference between MIMO and MU-MIMO?

Feature	MIMO	MU-MIMO
Introduced In	802.11n (Wi-Fi 4)	802.11ac Wave 2 (Wi-Fi 5) – downlink; 802.11ax (Wi-Fi 6) – uplink + downlink
How It Works	Multiple antennas on AP and client used to send multiple spatial streams to ONE device	Multiple antennas on AP used to send separate streams to MULTIPLE devices simultaneously
Users Served	One user at a time (single-user)	Multiple users at the same time
Streams	Up to 4 spatial streams (802.11n), 8 (802.11ac)	Up to 4 users simultaneously (802.11ac), 8 users (802.11ax)
Benefit	Increases throughput for a single device	Increases overall network capacity in multi-device environments
Limitation	Other clients must wait for current transmission to finish	Requires client capability and AP to perform beamforming and scheduling

10. What are PPDU, PLCP, and PMD in the PHY layer?

PPDU – Physical Protocol Data Unit

The PPDU is the complete frame format at the PHY layer. It is the definitive unit of data that is actually transmitted over the air. Every Wi-Fi transmission visible on the wireless medium exists as a complete PPDU.

A PPDU consists of three primary components:

- **PLCP Preamble:** Used for synchronization.
- **PLCP Header:** Contains metadata like data rate and frame length.
- **PSDU (the MAC data):** The actual payload being transported.

PLCP – Physical Layer Convergence Protocol

The PLCP is the upper sublayer of the PHY layer. It serves as an essential interface between the MAC layer (the logical portion of the network) and the physical medium itself.

Its key responsibilities include:

- **Framing:** Adding the preamble and header to the MAC data (MPDU) to encapsulate it into the final PPDU.
- **Rate Negotiation:** Signaling critical parameters—including data rate, channel width, and MCS (Modulation and Coding Scheme)—to the receiver using the PLCP header.
- **Clear Channel Assessment (CCA):** Monitoring the physical medium and reporting to the MAC layer whether the wireless channel is currently busy or free. This function is fundamental to the CSMA/CA (Carrier Sense Multiple Access with Collision Avoidance) protocol used in Wi-Fi.
- Provides a standardized interface, ensuring that the same MAC layer logic can operate with different underlying physical mediums.

PMD – Physical Medium Dependent

The PMD is the lower sublayer of the PHY layer. It handles the low-level technical aspects of physically transmitting and receiving individual bits over the radio medium.

Its key responsibilities include:

- **Modulation and Demodulation:** Converting digital data (bits) into analog radio signals for transmission, and converting received analog signals back into digital data.
- **Encoding:** Applying forward error correction (FEC) techniques.
- **Frequency and Power:** Controlling and managing the transmit frequency, power output levels, and all antenna operations.
- The PMD sublayer is the specific component that differs between various Wi-Fi standards. For example, it defines the difference between 802.11a (which uses OFDM at 5 GHz) and 802.11b (which uses DSSS at 2.4 GHz).

11. What are the types of PPDU? Explain the PPDU frame format across different Wi-Fi generations.

1. Non-HT PPDU (Legacy: 802.11a/g)

Used in the early days of OFDM, this format is simple and does not support modern features like MIMO (Multiple Input Multiple Output).

- **Structure:** L-STF -> L-LTF -> L-SIG -> Data
- **Key Fields:** It uses Legacy Short and Long Training Fields for synchronization and a single Signal field to define the data rate (up to 54 Mbps).
- **Usage:** Still used today for management frames (like Beacons) so that even the oldest devices can understand them.

2. HT PPDU (High Throughput: 802.11n / Wi-Fi 4)

This generation introduced MIMO and channel bonding. To remain backward compatible, it introduced the **Mixed Mode** format.

- **HT-Mixed Format:** L-STF -> L-LTF -> L-SIG -> **HT-SIG** -> HT-STF -> HT-LTF -> Data
- **HT-SIG Field:** This is the most important addition; it tells the receiver how many spatial streams are being used and if the channel is 20 MHz or 40 MHz.
- **HT-Greenfield:** A version that removed the "Legacy" (L-) fields for better efficiency, but it was rarely used because it would crash older devices on the same network.

3. VHT PPDU (Very High Throughput: 802.11ac / Wi-Fi 5)

Wi-Fi 5 focused on wider channels (80 MHz/160 MHz) and MU-MIMO.

- **Structure:** L-SIG -> **VHT-SIG-A** -> VHT-STF -> VHT-LTF -> **VHT-SIG-B** -> Data
- **VHT-SIG-A:** Contains information global to the whole frame (bandwidth, number of users).
- **VHT-SIG-B:** Contains information specific to each individual user when using Multi-User MIMO, such as their specific MCS (Modulation and Coding Scheme).

4. HE PPDU (High Efficiency: 802.11ax / Wi-Fi 6)

Wi-Fi 6 introduced **OFDMA**, requiring the most complex header structure yet to manage "Resource Units" (sub-channels).

- **Structure:** Includes a **Repeated L-SIG (RL-SIG)** which helps the receiver identify the frame as Wi-Fi 6 quickly.
- **HE-SIG-B:** This is the "map" for OFDMA. It tells Device A to look at one set of frequencies and Device B to look at another, allowing simultaneous data transmission.
- **Types:**
  - **HE SU:** For a single user.
  - **HE MU:** For multiple users (OFDMA/MU-MIMO).
  - **HE TB (Trigger-Based):** Used when the AP "triggers" multiple clients to talk back at the exact same time.

5. EHT PPDU (Extremely High Throughput: 802.11be / Wi-Fi 7)

Wi-Fi 7 introduces **320 MHz channels** and **4096-QAM**.

- **U-SIG (Universal SIG):** A new, highly compressed header designed to be future-proof for several generations.
- **EHT-SIG:** Provides the massive amount of signaling required for Multi-Link Operation (MLO), where a device can use 2.4, 5, and 6 GHz bands simultaneously.

12. How is the data rate calculated?

Data Rate = (Subcarriers × Bits/Symbol × Coding Rate × Spatial Streams) / Symbol Duration
Parameters Explained:

• Number of Data Subcarriers: Depends on channel width (e.g., 20 MHz OFDM → 48 data subcarriers; 40 MHz → 108 subcarriers).

• Bits per Symbol (Modulation): BPSK=1, QPSK=2, 16-QAM=4, 64-QAM=6, 256-QAM=8, 1024QAM=10.

• Coding Rate (FEC Rate): Forward Error Correction adds redundancy. Common rates: 1/2, 2/3, 3/4, 5/6. A rate of 3/4 means 75% of bits are data, 25% are error correction.

• Spatial Streams (MIMO): Each additional spatial stream multiplies the data rate (e.g., 2 streams = 2× rate).

• Symbol Duration: Typically 4 µs in 802.11a/g/n/ac (3.2 µs OFDM + 0.8 µs GI); 12.8 µs in 802.11ax (with 0.8/1.6/3.2 µs GI options). Example Calculation – 802.11n, 20 MHz, 64-QAM, 3/4 coding, 1 stream, short GI:

• Data subcarriers: 52

• Bits per subcarrier: 6 (64-QAM)

• Coding rate: 3/4

• Spatial streams: 1

• Symbol duration: 3.6 µs (with short GI)

• Data rate = (52 × 6 × 3/4 × 1) / 3.6 µs = 234 / 3.6 = 65 Mbps